

Artificial Intelligence Fundamentals

CSE4002

SUBJECT LEARNING GUIDE

Teaching Period Semester 1, 2025

Subject Coordinator

Dr. Yang Zhao

SUBJECT DETAILS

Subject Code:	CSE4002
Subject Title:	ARTIFICIAL INTELLIGENCE FUNDAMENTALS
Teaching Period:	Semester 1
Location(s):	Melbourne
Credit Points:	15
Delivery Mode:	DAY
AQF Level:	8

ENROLMENT REQUIREMENTS

Prerequisites:	CSE4PE
Co-requisites:	N/A
Assumed Skills & Knowledge:	N/A
Special Study Requirements:	N/A

Additional subject requirements: Internet access. All resources are available online via the LMS.

Class materials: All materials are available online via the LMS.

STAFF CONTACTS

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Consultation time/s:	Appointment by email

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ABOUT THIS SUBJECT

This subject covers the fundamental areas of Artificial Intelligence, focussing on knowledge representation and search. Main topics include historical foundations and applications areas; state-space search; game-playing; knowledge representation languages including predicate calculus, semantic networks, conceptual graphs and frames; rule-based expert systems; and an introduction to symbolic and connectionist machine learning.

SUBJECT INTENDED LEARNING OUTCOMES (SILOs)

Upon successful completion of this subject, you should be able to:

- 1 Devise appropriate representations for state space search and game playing to solve practical problems.
- 2 Represent knowledge using semantic networks, scripts, frames and logic-based mechanisms and write simple Prolog programs to enable automated reasoning.
- 3 Construct simple expert systems to solve a real-world problem.
- 4 Analyse and design basic machine learning algorithms to model a practical case.

GRADUATE CAPABILITIES (GC)

GC	GC Point
Communication	
Inquiry and Analysis	
Personal and Professional	
Discipline Knowledge and Skills	

LEARNING ACTIVITIES

View your personal timetable and the University timetable for all available subjects at [Timetables](#). You should also regularly check this subject's Learning Management System (LMS) site for updated information.

Session	Date	Learning Focus and Activities	Contact hours
1	Week 1	Introduction to AI	4
2	Week 2	Blind Search	4
3	Week 3	Heuristic Search	4
4	Week 4	Logic and Knowledge Representation	4
5	Week 5	Logic and Knowledge Representation	4
6	Week 6	Automated Reasoning	4
7	Week 7	Expert Systems	4

Session	Date	Learning Focus and Activities	Contact hours
8	Week 8	Machine Learning	4
9	Week 9	Machine Learning	4
10	Week 10	Game-Playing	4
11	Week 11	AZURE: A Cloud-Based AI Platform	4
12	Week 12	Revision	4
		Total	48

Details of any self-allocation activities
Students are required to self-allocate using the Allocate+ system.

SELF-DIRECTED STUDY

In addition to the learning activities specified, you are expected to allocate approximately **<number; see [Staff Guidelines for Calibrating Student Workload](#)> hours per week of self-directed study in this subject.** This time includes subject preparation, completing learning activities, readings, assessment preparation and other related work to enhance your participation in this subject and your learning at La Trobe.

LEARNING RESOURCES & TECHNOLOGIES

Resource Type	Title	Author and Year	Publisher
Required			
book	Artificial intelligence: structures and strategies for complex problem solving (6th ed).	Luger, G. 2009	Addison-Wesley
Recommended			
book	Artificial Intelligence: a modern approach (3 rd ed)	Russel, S. and Norvig, P. 2016	Prentice-Hall

ASSESSMENT AND FEEDBACK SUMMARY

Note: all assessment instructions and rubrics with detailed criteria and level of performance are found on the Learning Management System two weeks prior to classes commencing.

Week/ Session	Date due	%	Assessment type	Key Criteria	Feedback method	SILOs assessed
1.	LMS	20	Case Study	A practical assignment to apply AI theory and technology to solve a real-world problem.	Written feedback	SILO 1 and 3
2.	LMS	30	5 Quizzes			SILOs 2 and 4
3.	LMS	50	Final Exam	Assess all knowledge learned in this subject.		SILOs 1, 2, 3 and 4

Additional information about assessment & feedback

Students should keep a copy of all assignments submitted for assessment.

MANDATORY REQUIREMENTS

There are a variety of assessment hurdles or conditions for passing a subject other than the overall mark. **Mandatory Requirements** may be built into the assessment of the learning outcomes of your subject.

There are no Hurdle Requirements for this subject / The Hurdle Requirements for this subject are:

Other mandatory requirements for this subject are:

GENERAL ASSESSMENT REQUIREMENTS

SUBMITTING WRITTEN WORK WITH TURNITIN

Turnitin is a web-based text-matching software system used at La Trobe University to assist you in writing your assignments and checking for similarity with existing published work. Please remember to allow adequate time to submit your assignment to Turnitin. **It is your responsibility to submit your assessment by the due date.** Not receiving an originality report is not an acceptable reason for requesting an assignment extension unless it is a required part of the assessment.

POLICIES, PROCEDURES AND GUIDELINES

The University has a comprehensive policy framework to which both staff and students must adhere. You should familiarise yourself with those policies, procedures and guidelines likely to affect you, especially the following:

- [Academic Progression Review Policy](#)
- [Assessment Policy](#)
- [Health and Safety Policy](#) (OHS)
- [Privacy Policy](#)
- [Student Academic Misconduct Policy](#)

ADJUSTMENTS TO ASSESSMENT

You may be affected by a range of adverse circumstances while you are preparing for or undertaking an assessment. There are avenues for adjustments to your assessment depending on the circumstances you face, including opportunities to apply for extensions to submission dates, Special Consideration and Replacement Examination Dates. For more detailed information, refer to the [Assessment Adjustments](#) website and the [Assessment Procedure – Adjustments \(including Special Consideration\)](#).

REQUEST AN EXTENSION OF TIME TO SUBMIT AN ASSIGNMENT ASSESSMENT

Where you know in advance that you will not be able to submit an assessment by the due date due to adverse circumstances that have affected you during the preparation of the assessment, you will need to [request a Short Extension of Time](#).

Applications for Short Extension of Time must be submitted no later than 11.59pm (Melbourne time) on the due date of the assessment.

An extension of three (3) calendar days is automatically approved, and no evidence is required (with the exception of Learning Access Plans). After you submit your form, you will receive your new date in a few minutes.

Short Extension of Time can only be granted *once* for each assessment for students who do not have a LAP with relevant provisions.

Penalties normally apply if you submit an assessment after the due date. **Poor time management is not an acceptable reason for an extension.**

To apply, go to [Request for an extension of time to submit an assessment task](#).

SPECIAL CONSIDERATION

If you have experienced **acute, significant and unexpected circumstances** that substantially affect your ability to complete an assessment to the best of your potential, you may be eligible to apply for **Special Consideration**. To do this or find more information, visit the [Assessment Adjustments](#) webpage.

STUDENTS WITH A LEARNING ACCESS PLAN (LAP)

Students who have a Learning Access Plan (LAP) must apply for any adjustments they may need via the same channels but may have access to more flexible arrangements depending on their circumstances.

If you have an LAP and need an extension, you should still apply using the Assessment Adjustment form within the designated timeframe. Please attach your LAP to the Assessment Adjustment form as your supporting evidence.

If your LAP outlines that you may require an extension of more than 7 calendar days, please submit a Short Extension of Time request (for the initial 7 days) and a Special Consideration application (for any additional days as outlined in your LAP).

If your existing circumstances exacerbate, or a new circumstance that is not stipulated in your LAP occurs, you should submit a new Special Consideration application.

Please refer to the [Assessment Procedure – Adjustments](#) and your LAP.

REVIEW AND RE-MARK

For information on what to do if you wish to seek a review of an assessment outcome, [see Part B](#) of the Validation and Moderation Procedure.

ACADEMIC INTEGRITY

Academic integrity means being honest in academic work and taking responsibility for learning the conventions of scholarship. Academic integrity education is integral to the learning experience at La Trobe University. The University requires its academic staff and students to observe the highest ethical standards in all aspects of academic work, and it demonstrates its commitment to these values by awarding due credit for honestly conducted scholarly work, and by penalising academic misconduct and all forms of cheating.

The penalty for submitting an assessment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to **Academic Integrity – Schedule of Penalties and Actions** within the [Student Academic Misconduct Policy](#).

You should understand and be able to practise integrity in all your academic work, be familiar with the [Academic Integrity website](#) and complete the **Academic Integrity Module (AIM)** found in your LMS.

If you have any specific questions about **academic integrity** and your assessment, your Subject or Course Coordinator will be able to assist.

STUDENT FEEDBACK ON SUBJECT SURVEYS

The **Student Feedback on Subjects (SFS) Survey** is one of the quality assurance processes that occur across the University. In this survey, you can tell us about your learning experiences in this subject. We value your feedback. Your views are taken seriously and assist us to improve the learning experience for future students through the La Trobe University Subject Monitoring process.

The surveys will be available for you to complete towards the end of the subject teaching period. Feedback from previous students who completed this subject is available in the LMS site for this subject.

For more information, you may contact surveys@latrobe.edu.au.

CHARTER OF STUDENT RIGHTS AND RESPONSIBILITIES

As a La Trobe University coursework student, you have rights and responsibilities that are governed by La Trobe statutes and regulations, as well as policies and procedures. La Trobe University aspires to provide

an outstanding student experience that is conducive to scholarly activity and supportive of your learning goals.

The **Charter of Student Rights and Responsibilities** is a summary of your rights and responsibilities as a student at La Trobe. The Charter was developed in consultation with La Trobe students.

STUDENT COMPLAINTS

The **Student Complaints Management System (SCM)** is a formal mechanism for students to lodge concerns about any aspect of their university experience. A student complaint may be about an administrative matter, a problem with a subject or course, or the conduct of a student or staff member.

For more information about the **Student Complaint process** and how to lodge a complaint, visit the [Complaints and Feedback](#) page.

GETTING HELP AND SUPPORT AT UNI

A range of services at La Trobe can assist you with your studies, health and wellbeing. For the full range of services and facilities available for you, go to **Student Help and Support**.

LIBRARY LEARNING SERVICES

Your University Library offers more than a place to access information or a quiet place to study. We have a team of friendly specialists and a wide range of services available to assist you in your academic studies. Discover how by visiting your [Library website](#) and the [Study and Research Support](#) webpage, and exploring the following options:

- [ACHIEVE@UNI](#)
Browse a collection of online learning resources, including writing samples from real students, to help you understand and complete your assessments.
- [PATHFINDER – YOUR GUIDE TO LEARNING SUCCESS](#)
Find practical online advice to help you develop effective study habits, improve your assessments, prepare for exams, and enhance your research skills. Also, learn how to connect with friendly specialists for assistance, including Academic Skills Language Advisors, Librarians, Peer Learning Advisors, and Maths, Science and Coding Hub Tutors.
- [ENGLISH FOR SUCCESS – ACADEMIC ENGLISH TRANSITION](#)
Find strategies to help you transition to study in Academic English, whether you are an international student or a domestic student in your first year of study.
- [STUDIOSITY](#)
Studiosity is an online writing feedback and study skill chat service that's available when you need it. Access this service through your Subject LMS page, or the Library's Study and Research Support webpage.
- [ASSESSMENT PLANNER](#)
Create a personal assessment schedule.
- [EXPERT SUBJECT HELP GUIDES](#)
Find expert discipline and specialist help for your field of study.
- [REFERENCING AT LA TROBE UNIVERSITY](#)
Understand why, when and how to reference your work.

- [ACADEMIC REFERENCING TOOL](#)
Get detailed examples of referencing styles.
- [CHAT TO LIBRARY STAFF](#)
Chat online with a specialist – not a chatbot – for advice on finding information, referencing and more.
- [CONTACT US](#)
Call or email us to learn how we can assist you.

OTHER LEARNING AND SUPPORT PROGRAMS

There are many resources, including services, to help you with your academic studies at La Trobe. For more information, visit the [Learning and Support Programs](#) page.

ASK LA TROBE

ASK La Trobe is a service open to all La Trobe students who need guidance with administrative matters during their time at La Trobe. ASK La Trobe can provide you with help relating to enrolments and timetabling, fees and financial support, exams, results and graduations, and student life and support services.

Go to [ASK La Trobe – Current Students](#) and start by looking at our **FAQs** or visit us at your campus **ASK La Trobe Help Zone**.

If you can't find what you're looking for:

- **View the FAQs**
- **Call 1300 La Trobe (1300 528 762)**
- **Ask a question [online](#)**
- **Visit the ASK La Trobe Help Zone.** Join the queue for the Bundoora, Bendigo or city campuses using the **QLess: Virtual Queuing system**.

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HEALTH AND WELLBEING

Your health and wellbeing are an important part of being a successful student. Studying can be stressful, and many factors can affect your mental and physical health, in and outside of the University. Check out the services available to you at La Trobe to support your [health and wellbeing](#).